

INTRODUCTION

The urgent transition from fossil fuels to renewable energy sources has created a pressing need for innovative methods to measure and predict electricity generation. In 2017, fossil fuels accounted for 81% of global energy consumption, emphasizing the importance of integrating solar and wind energy into the energy mix. This shift is driven by growing concerns about climate change and rising energy costs. Key factors influencing solar and wind power generation include solar irradiance, humidity, and wind speed, which are critical for effective project planning and resource management. This work aims to propose a method for predicting wind and solar energy using machine learning algorithms, focusing on data obtained from "Bhadla solar plant rajasthan India". By employing the Random Forest algorithm, we achieve accurate forecasts of solar radiation and wind speed, evaluated through metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE).

A. Data collection for solar plant

Many variables were taken into consideration but the ones that were chosen, given the previous analysis, were temperature, pressure, humidity, wind direction, wind speed. Most of the already mentioned parameters were obtained from "Bhadla solar plant rajasthan India" which gave us the data collected from three different weather stations distributed strategically throughout the PV plant, located in Rajasthan, India (27.5176° N, 71.9304° E). Other factors such as pressure, humidity, temperature and wind direction were collected in an average-value form during a measurement period of six months



Data collection for wind plant

This dataset is a comprehensive collection of meteorological observations and wind power generation data, recorded hourly starting from January 2, 2017, at one of the company's operational wind turbine sites. It includes detailed weather variables such as temperature, humidity, dew point, and wind characteristics, along with power generation data from the wind turbines.

dataset contain temperature_2m, relativehumidity_2m, dewpoint_2m, windspeed_10m, winddirection_10m, windgusts_10m, power

B. Pre-processing

To make the database reading more agile, the same format was given to each of the columns that stored the variables involved in the study. For this, the average measurements per hour of irradiance, wind speed and direction, temperature, and humidity. The measurement units for each parameter are shown in Table I.

TABLE I
UNITS USED IN THE STUDY

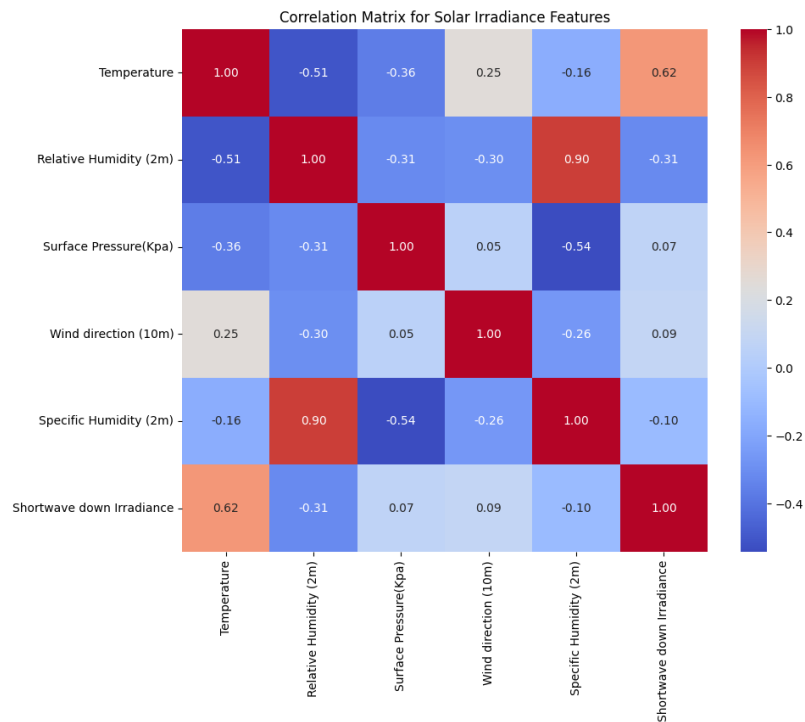
<i>Values</i>	<i>Units</i>
Wind Velocity	m/s
Wind Direction	°(Degrees)
Temperature	°C
Pressure	mmHg
Irradiance	W/m ²
Humidity	%

We don't change anything in wind power plant data.

C. Data analysis

1. Irradiance for Solar Plant:

- **Solar Irradiance** is the amount of solar power received per unit area at a given time, typically measured in W/m² (Watts per square meter). This is a crucial factor for predicting solar power generation.
- In this dataset, solar irradiance can be calculated or obtained from additional data sources (e.g., using pyranometer readings or satellite-based data).
- The relationship between solar irradiance and temperature, humidity, and other weather factors can help predict energy output for solar power plants.



2. Power for Wind Energy Plant:

- **Wind Power** generation is directly influenced by meteorological factors such as **wind speed**, **wind direction**, **temperature**, and **pressure**. These variables are already part of the dataset.
- Wind power output can be modeled using the **power curve** of wind turbines, where power is a function of wind speed (and other factors like wind gusts and direction).
- In this dataset, **power generation for wind energy** is already recorded as the target variable ("Power"). This can be used for predictive modeling and optimization of wind turbine operations.



D. RESULTS

The ML algorithms used for this study were programmed in the 3.7 version of Python. For reading and manipulating purposes of the obtained datasets, the Pandas library was used; for the graphic analysis of information like the Correlation Matrices, Heat-Map and plotting in general, Pyplot and Color libraries from matplotlib were used. In addition, the sklearn library was used for the implementation of all the ML algorithms.

For each of the algorithms, 25% of both datasets, Wind and Solar, was used for testing. To analyze our proposal, the experiments were carried out with four classic ML algorithms: (i) Multi-layer Perceptron, (ii) Support Vector Machine, (iii) Random Forest Their individual performance was based on two accuracy-indicators; MSE as shown in equation (1) and MAE as shown in equation (2).

For Solar Power Prediction:

- **Random Forest:**
 - **Mean Squared Error (MSE):** 2830.74
 - **Mean Absolute Error (MAE):** 24.65
 - The Random Forest model has the lowest MSE and MAE, indicating a better fit and higher accuracy for solar power prediction.
- **Support Vector Regression (SVR):**
 - **Mean Squared Error (MSE):** 7187.42
 - **Mean Absolute Error (MAE):** 51.87
 - SVR performs significantly worse than Random Forest, with much higher error metrics, indicating it is less effective for this task.
- **Multi-Layer Perceptron (MLP):**
 - **Mean Squared Error (MSE):** 2871.19
 - **Mean Absolute Error (MAE):** 30.43
 - **R² Score:** 0.9747
 - MLP provides a good fit with a high R² score, showing it explains a substantial amount of the variance in solar power prediction, but has higher MSE and MAE compared to Random Forest.

For Wind Power Prediction:

- **Random Forest:**
 - **Mean Squared Error (MSE):** 0.0263
 - **Mean Absolute Error (MAE):** 0.1226
 - **R² Score:** 0.6843
 - Random Forest performs well with a low MSE, low MAE, and a decent R² score, indicating that the model is able to explain some of the variance in wind power predictions.
- **Support Vector Machine (SVM):**
 - **Mean Squared Error (MSE):** 0.0325
 - **Mean Absolute Error (MAE):** 0.1404
 - **R² Score:** 0.6106
 - SVM is slightly less accurate than Random Forest, with a higher MSE and MAE. The R² score is also lower, indicating a lesser ability to explain the variance in the data.
- **Multi-Layer Perceptron (MLP):**
 - **Mean Squared Error (MSE):** 0.0948
 - **Mean Absolute Error (MAE):** 0.2542
 - **R² Score:** -0.1371
 - MLP performs the worst for wind power prediction, with the highest MSE and MAE. The negative R² score indicates that the model performs worse than simply predicting the mean value, which suggests poor model performance.